EE312

Experiment 5
The Operational Amplifier As an Amplifier and a Comparator

Eng. Abdulaziz Alqahtani

Experiment 5 Objectives

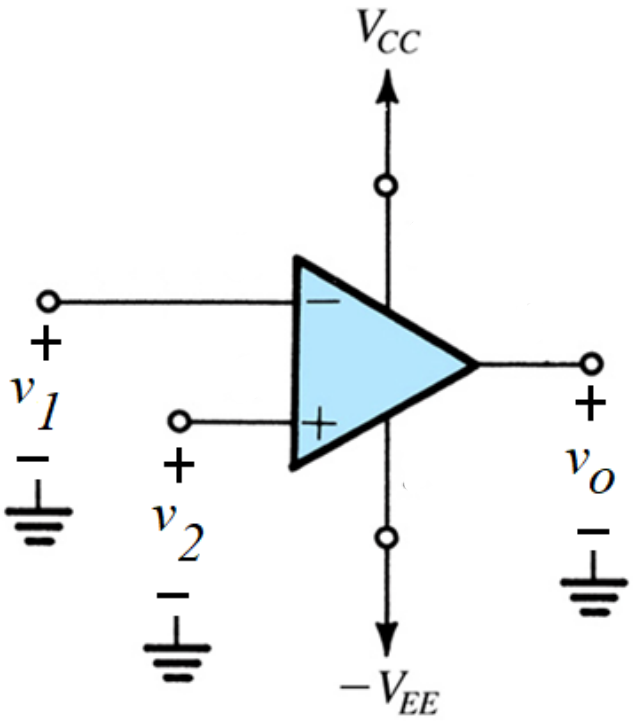
- To be acquainted with the essence of operation of the operational amplifier (op amp).
- To observe the difference between ideal and real-life op amp
- To build and test a non-inverting amplifier, measure its closed-loop voltage gain and bandwidth, and see the output voltage swing limitation.
- To build and test a voltage comparator employing limiter at its output

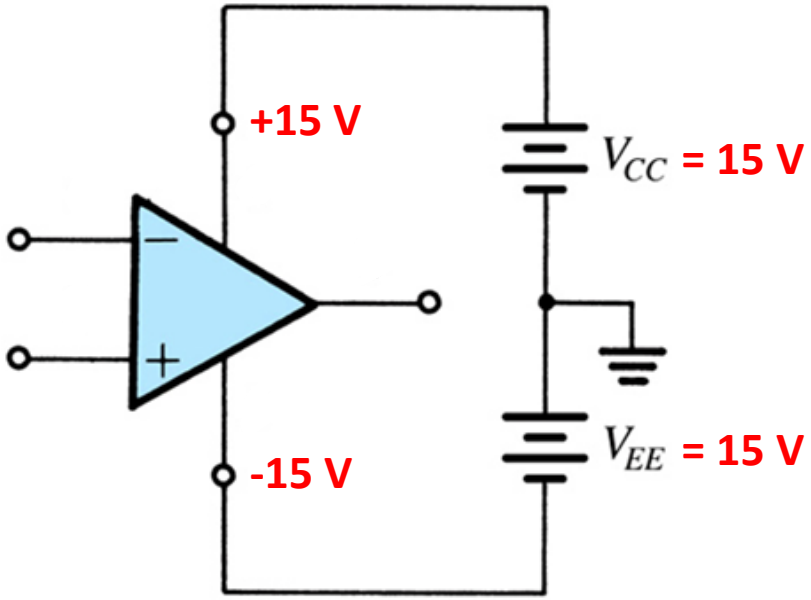
Operational Amplifier (op amp)

- The operational amplifier is a versatile *integrated circuit (IC)* that performs various types of analog signal processing tasks.

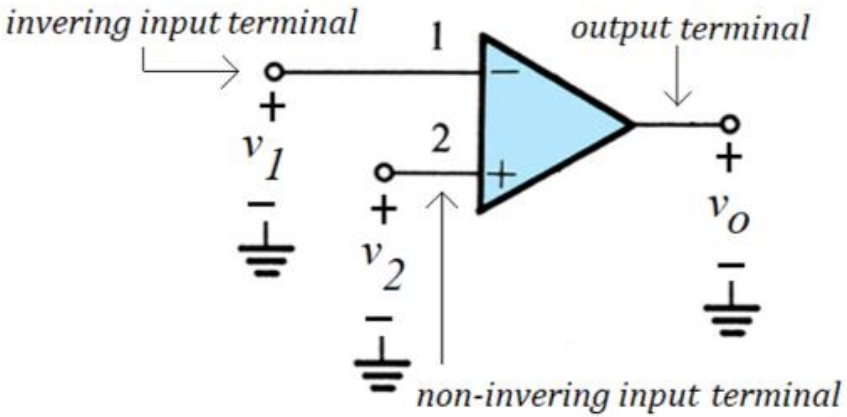
- The most popular op amp is the **µA741**, which was introduced by Fairchild in 1968 to become the industry standard.

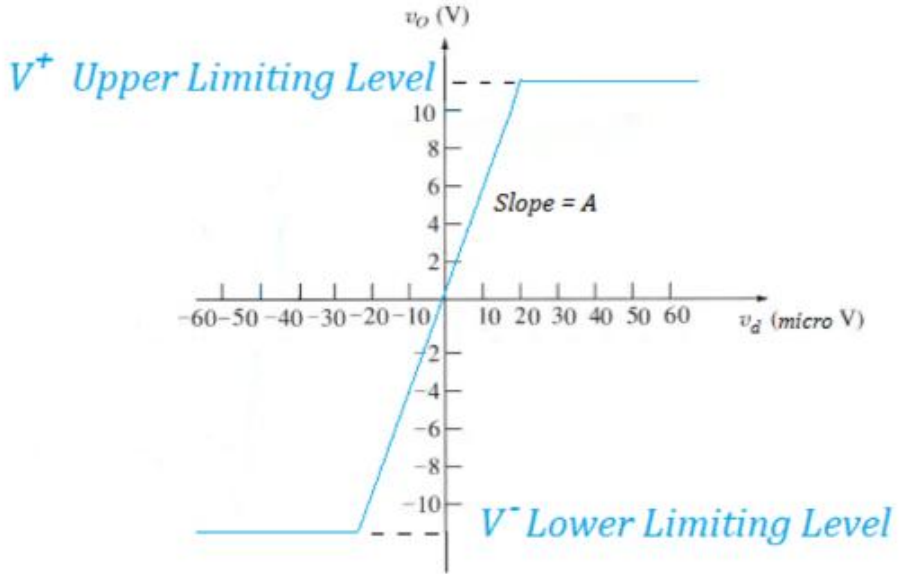
Symbol and Power Connections





Op Amp Operation





$$v_o = A\,(v_2 - v_1)$$

A is the open-loop gain

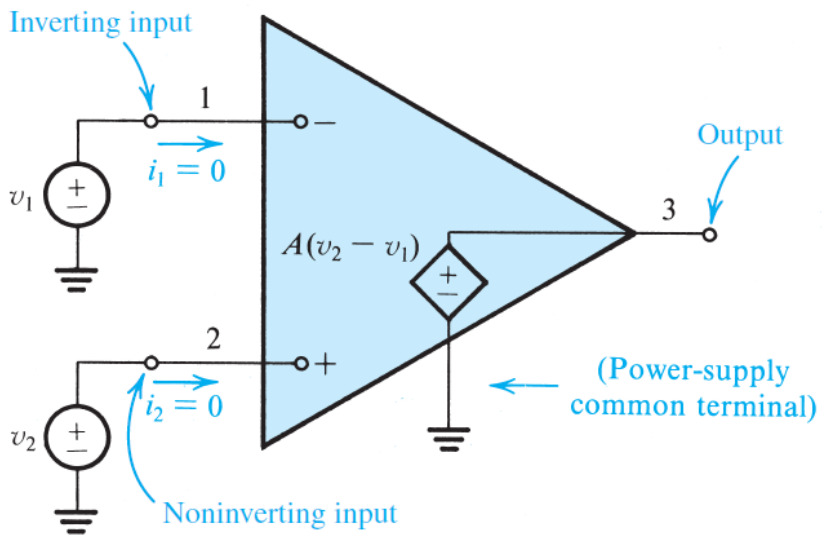
v_o is single-ended output

$$v_d = (v_2 - v_1)$$

v_d is the differential input

Ideal Op Amp

Equivalent Circuit

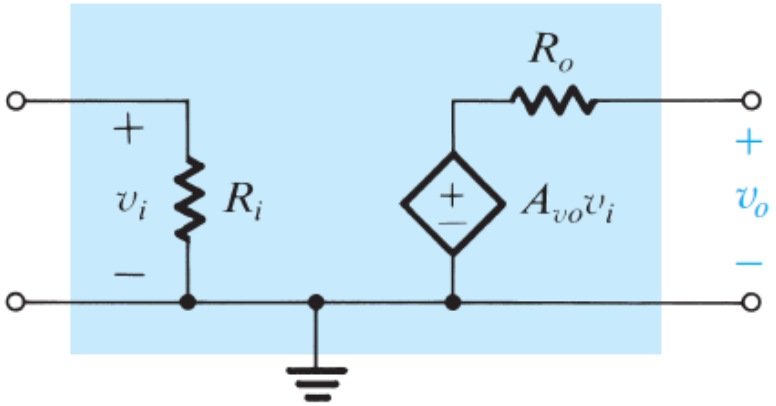


Characteristics

- **Infinite input Impedance**
- **Zero output impedance**
- **Infinite common-mode rejection**
- **Infinite open-loop gain A**
- **Infinite bandwidth**

Typical Op Amp

Equivalent Circuit

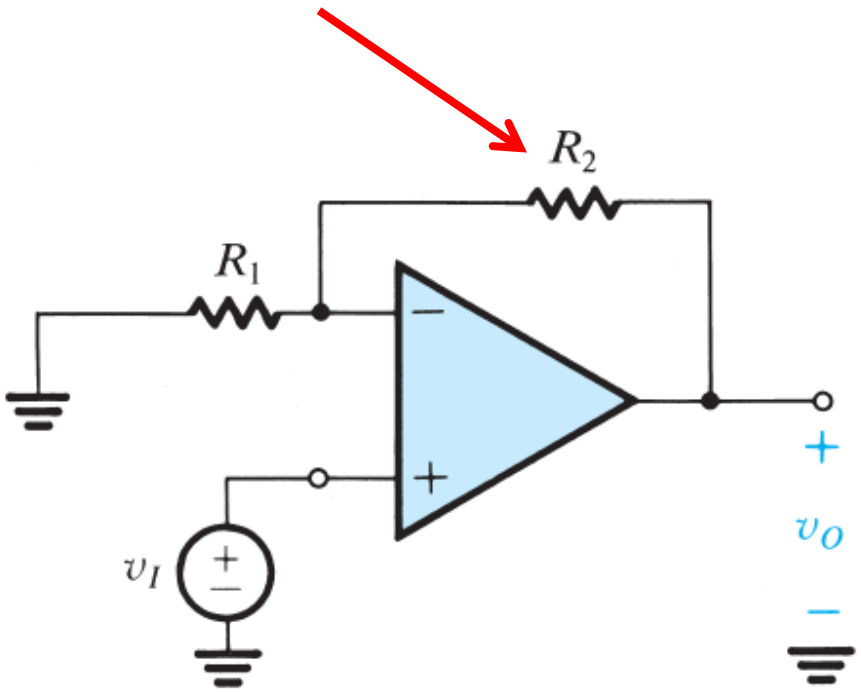


Characteristics

- Very large input impedance R_i
- Very small output impedance R_o
- Very large common-mode rejection
- Very large open-loop gain A
- Very large bandwidth

The Non-inverting Configuration

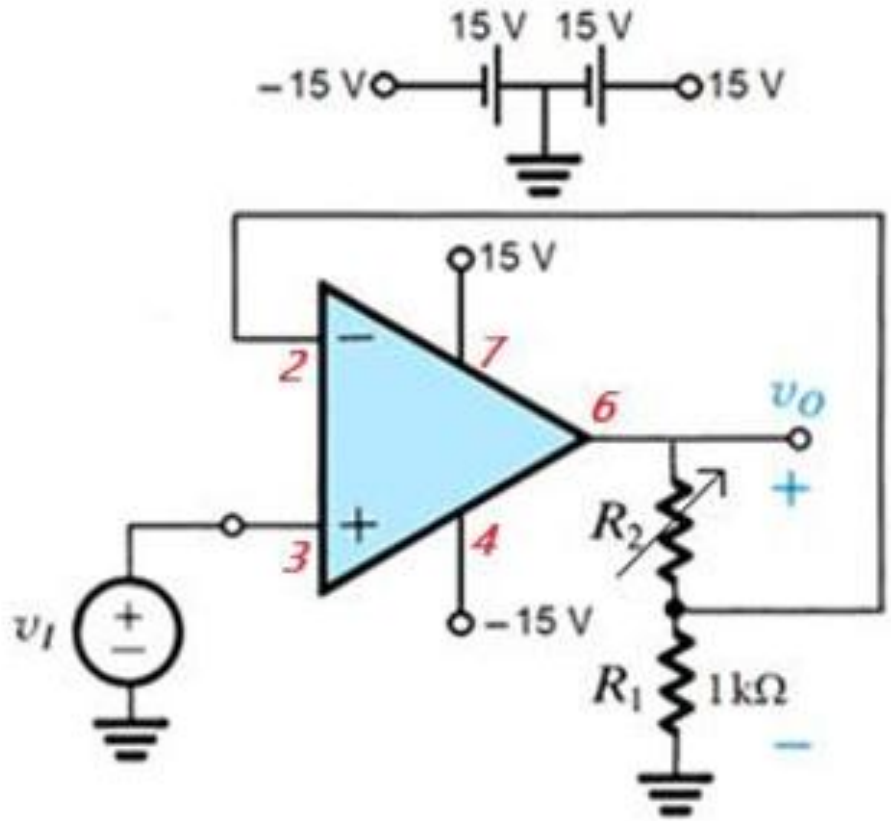
Negative feedback



$$G \equiv \frac{v_o}{v_i} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R2}{R1}$$

G is the closed loop gain, (Assuming infinite open-loop gain A)

The Non-inverting Amplifier



Op Amp as a Comparator

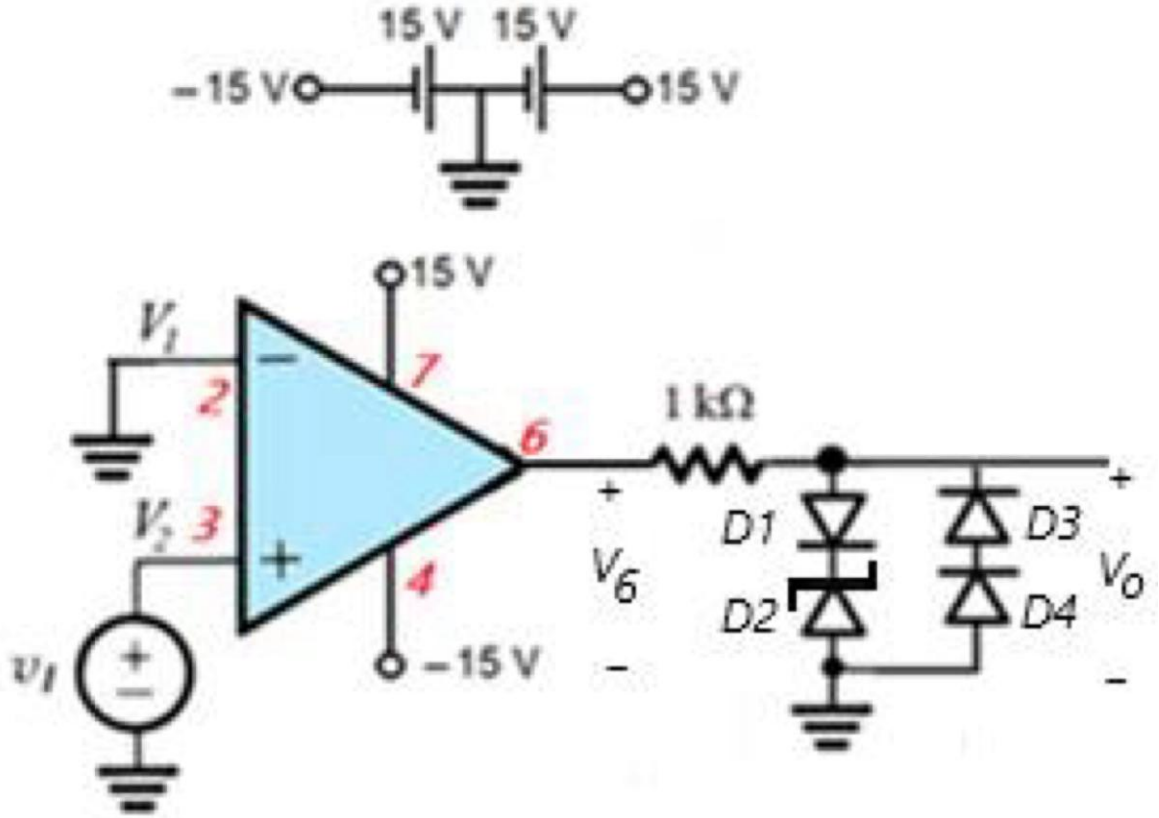
- If no negative feedback is employed, an op amp is said to operate in the **open-loop** mode.

- In that mode, an op amp basically works as a comparator.

- If $v_2 > v_1$ by more than few tens of microvolts then v_O = V_O^+ approximately V_{CC}

- If $v_1 > v_2$ by more than few tens of microvolts then v_O = V_O^- approximately $-V_{EE}$

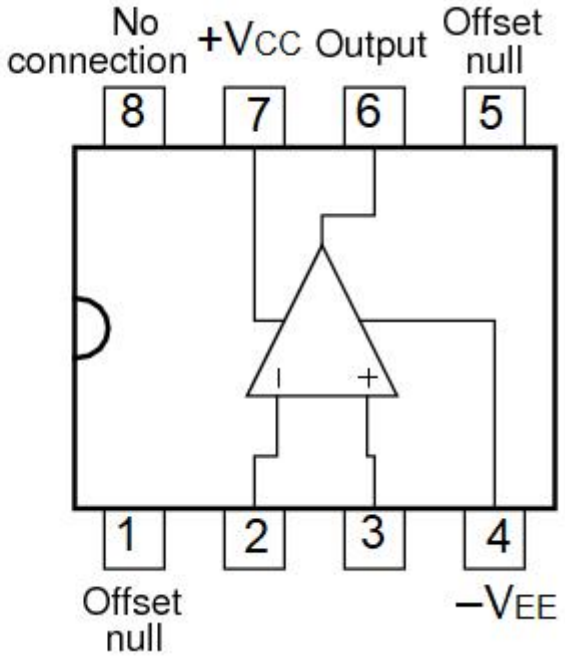
Comparator with Limited Output Range



Experimental Part

UA741 Op Amp

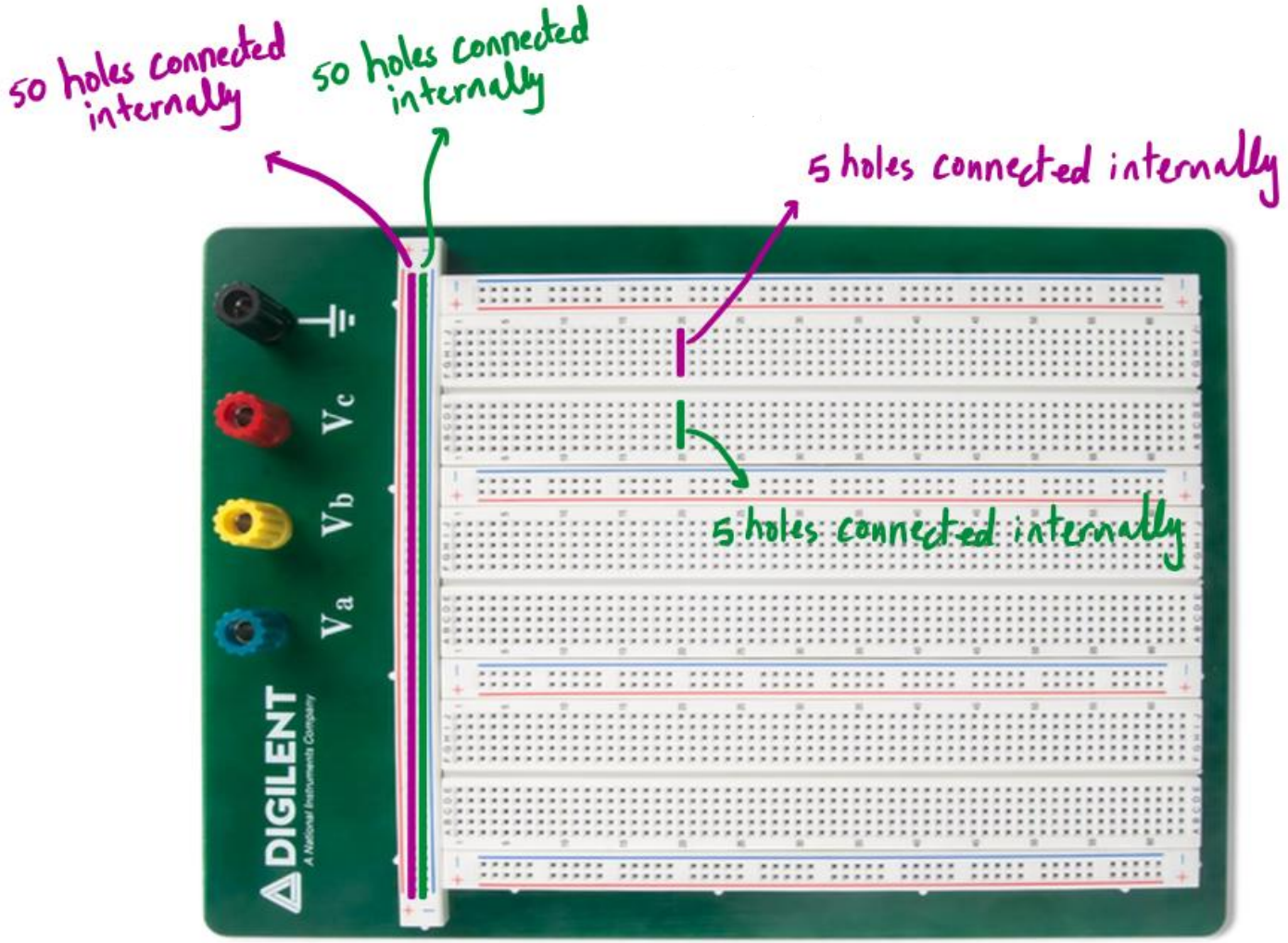




IC Package

Pin Diagram

Breadboard


50 holes connected internally
50 holes connected internally
5 holes connected internally
5 holes connected internally
Va
Vb
Vc
DIGILENT

Test Leads



BNC to Mini Grabber Hooks Cable
A. BNC Coaxial Connector
B. Mini grabber Hook
C. Ground Mini grabber Hook



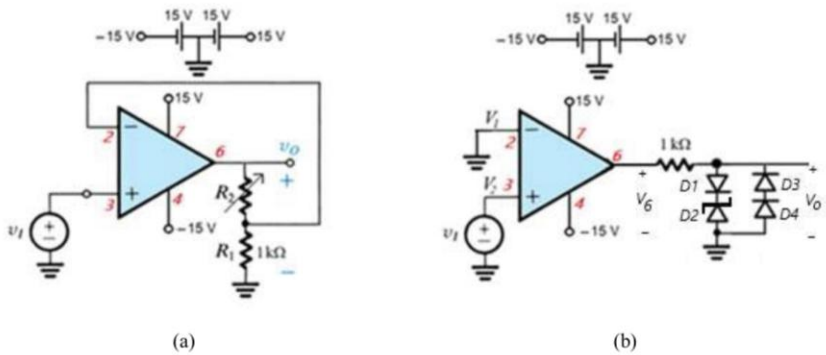


BNC to Double Alligator Clips Cable
A. BNC Coaxial Connector
F. Alligator Clip
G. Ground Alligator Clip

BNC to Double Banana Plugs Cable
A. BNC Coaxial Connector
D. Banan Plug
E. Ground Banan Plug

Appendix

PROCEDURE A: Non-inverting Amplifier



(a) (b)

Figure 5.4 **(a)** The noninverting amplifier. **(b)** Comparator with limited output range.

1. Construct the circuit shown in Fig. 5.4a; start off with a value of 10 kΩ for R_2.
2. Using a waveform generator, apply a 1-kHz, 400 $mV_{p\text{-}p}$ sinusoid to the noninverting terminal of your op amp.
3. Turn on your scope; wait for few seconds, press the **Default** key, and then the **Autoset** key. Display $v_I(t)$ and $v_O(t)$ on **CH1** and **CH2** respectively.
4. Use the **SCALE** knob of **CH1** to set its vertical sensitivity at **200 mV/div**.
5. Use the **SCALE** knob of **CH2** to set its vertical sensitivity at **1 V/div**.
6. Press the **Measure** key on your scope → **Add Measurement** → Activate the **Source 1** submenu → Use the **VARIABLE** knob to highlight **CH2** then press **Select** → Activate the **V/I** submenu → Highlight the appropriate item (**Pk-Pk, Max.** and **Min.**), then press the **Select** key to add anyone of them.
7. Switch the dc power supply, then the waveform generator on.
8. Record $v_I(t)$ and $v_O(t)$ on Fig. 5.5.
9. Record the peak-to-peak amplitude for the input and output voltage waveforms, then calculate the voltage gain G:

$$G = \frac{V_{o(\text{pk-pk})}}{V_{i(\text{pk-pk})}} \qquad (5.2)$$

10. Increase the frequency of the input signal until the peak amplitude of the output falls to 0.7 of its value at 1 KHz. Take note of that frequency from the **waveform generator reading**. Record that frequency (the bandwidth BW) in Fig. 5.5.
11. Reduce the frequency of the input signal down to 1 kHz.
12. Increase the scale of **CH2** to **5 V/div**. Replace the 10-kΩ resistor with a 100-kΩ one, then sketch the resulting waveforms of $v_I(t)$ and $v_O(t)$ on Fig. 5.6.
13. Obtain the *upper limiting level* V_O^+ and the *lower limiting level* V_O^- for $v_O(t)$ from the scope Max. and Min. measurements. Record those values in Fig. 5.6.
14. Swich the waveform generator then the the power supply off.

PROCEDURE B: Implementing a Comparator Using an OP-AMP

1. Assemble the circuit shown in Fig. 5.4b.
2. Use the same input signal that you used for part A.
3. Connect the non-inverting input of the op amp to display to **CH1** of the scope, the op-amp output to **CH2**, and the overall circuit output to **CH3**.
4. Add a **Max.** and **Min.** measurement for **CH3** of the scope.
5. Set the the sacle of **CH3** at 2 V/Div.
6. Swich the power supply, and then the waveform generator on.
7. Record $v_I(t)$, $v_6(t)$, and $v_O(t)$ on Fig. 5.7.
8. Obtain the upper and lower limiting levels for $v_6(t)$, and $v_O(t)$ from the oscilloscop Max. and Min. values, and then record them in lower part of Fig. 5.7.
9. Swich off the waveform generator then the the power supply off.

RESULTS

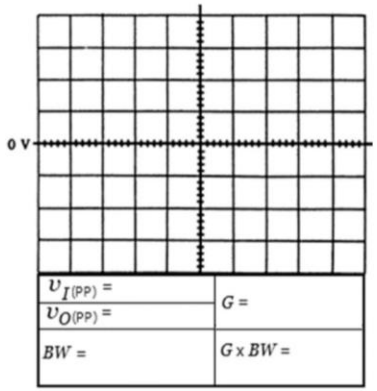


Figure 5.5

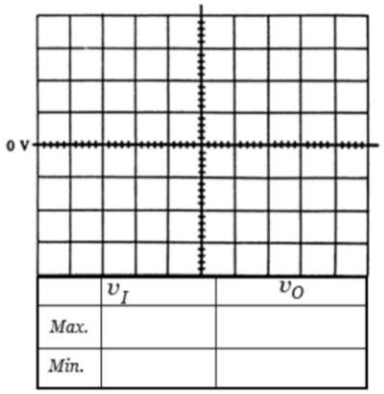


Figure 5.6

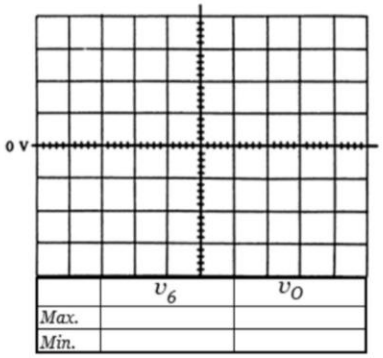


Figure 5.7

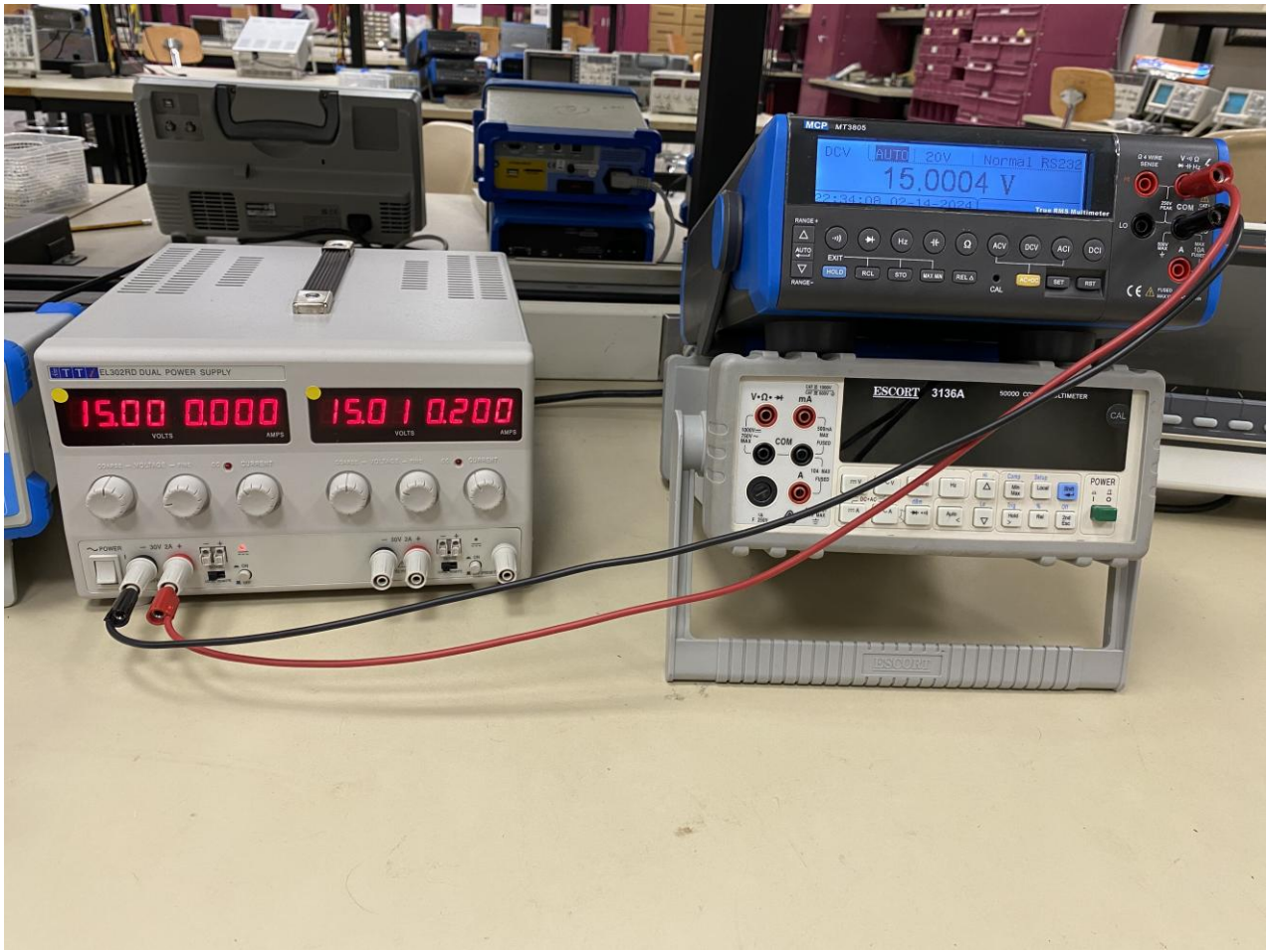
POST LAB QUESTIONS (PLQs)

1) *Procedure A*: Calculate the closed loop gain for the R_2 = 10 kΩ case using equation 5.2 and compare it to theoretical gain given by equation 5.1. Comment on the cause of the difference.
2) *Procedure A*: Calculate the gain bandwidth product for the R_2 = 10 kΩ case.
3) *Procedure A*:
   - State the cause of clipping (limiting) in the output voltage waveform resulting from increasing R_2 to 100 kΩ.
   - How are the limiting levels of $v_O(t)$ related to the power rails ± 15 V?
4) *Procedure A*: Compare the op amp you have used in the lab with an ideal op amp model in terms of bandwidth and output-voltage swing range.
5) *Procedure B*: Justify the value of the upper limiting levels of $v_6(t)$ and $v_O(t)$.
6) *Procedure B*: Justify the value of the lower limiting levels of $v_6(t)$ and $v_O(t)$.

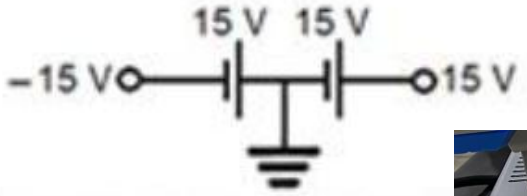
Setting First Power Rail to 15 V

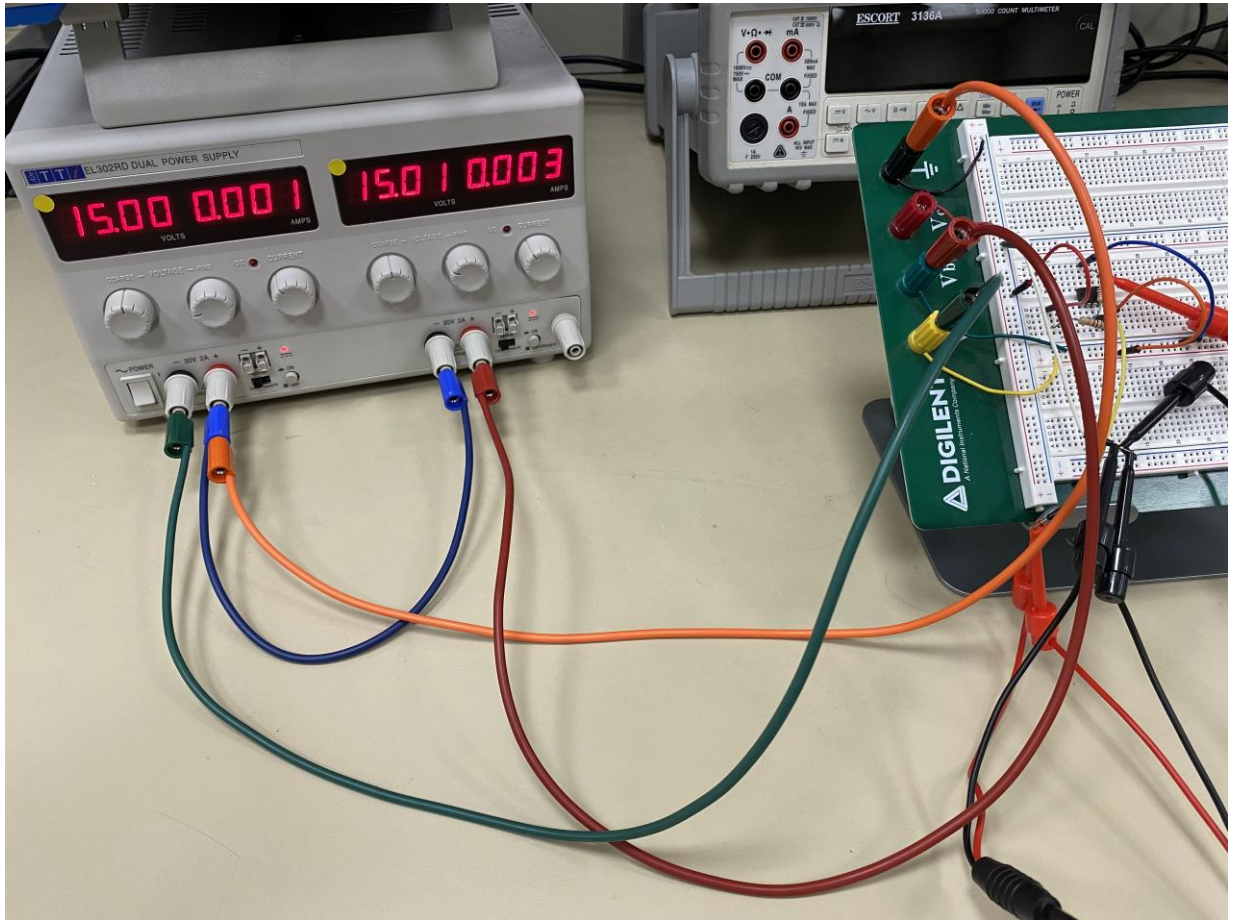


Setting Second Power Rail to 15 V

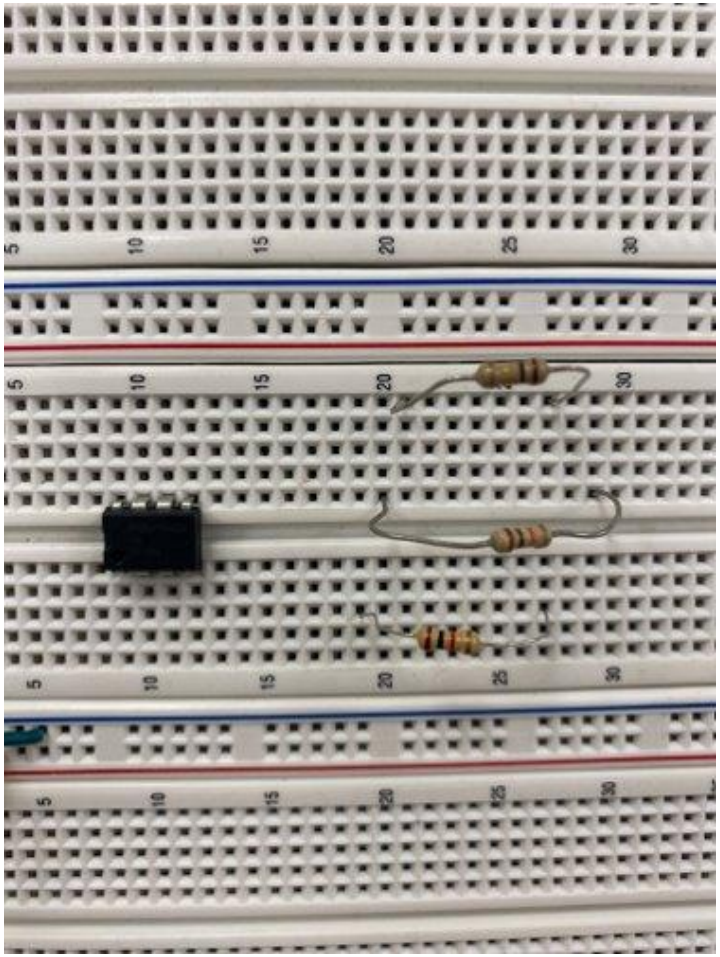


Power Rails Connection with Breadboard

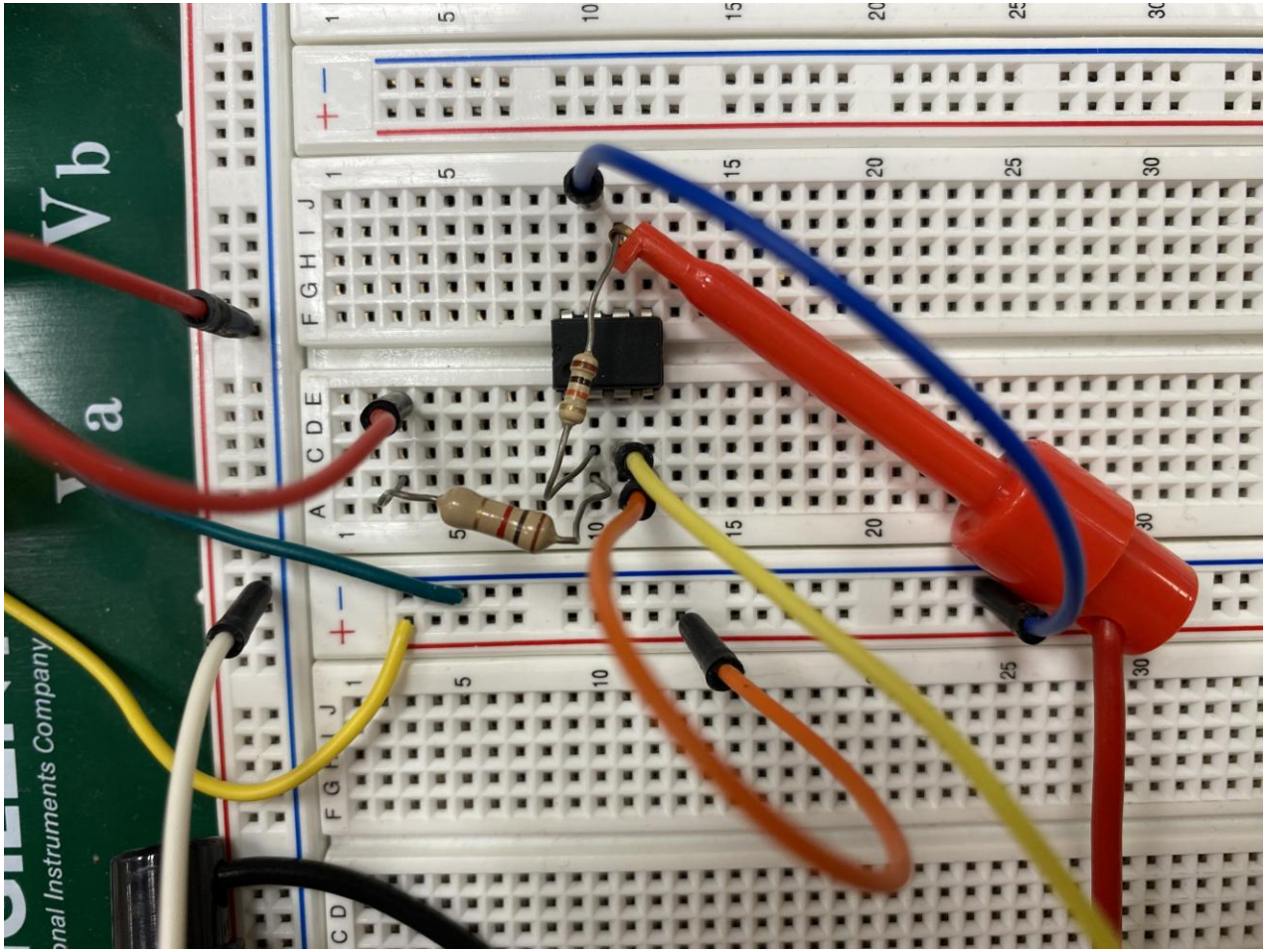


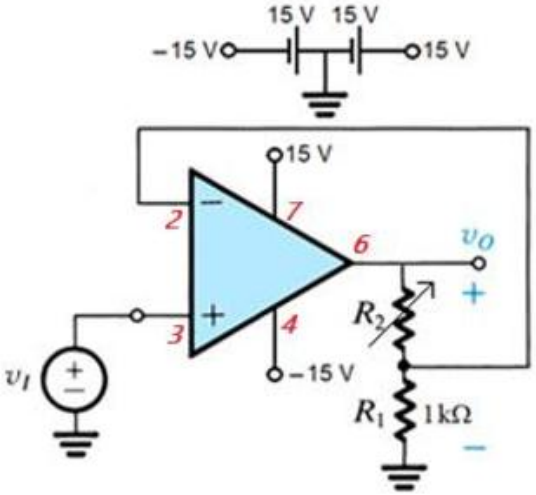


Procedure A Materials

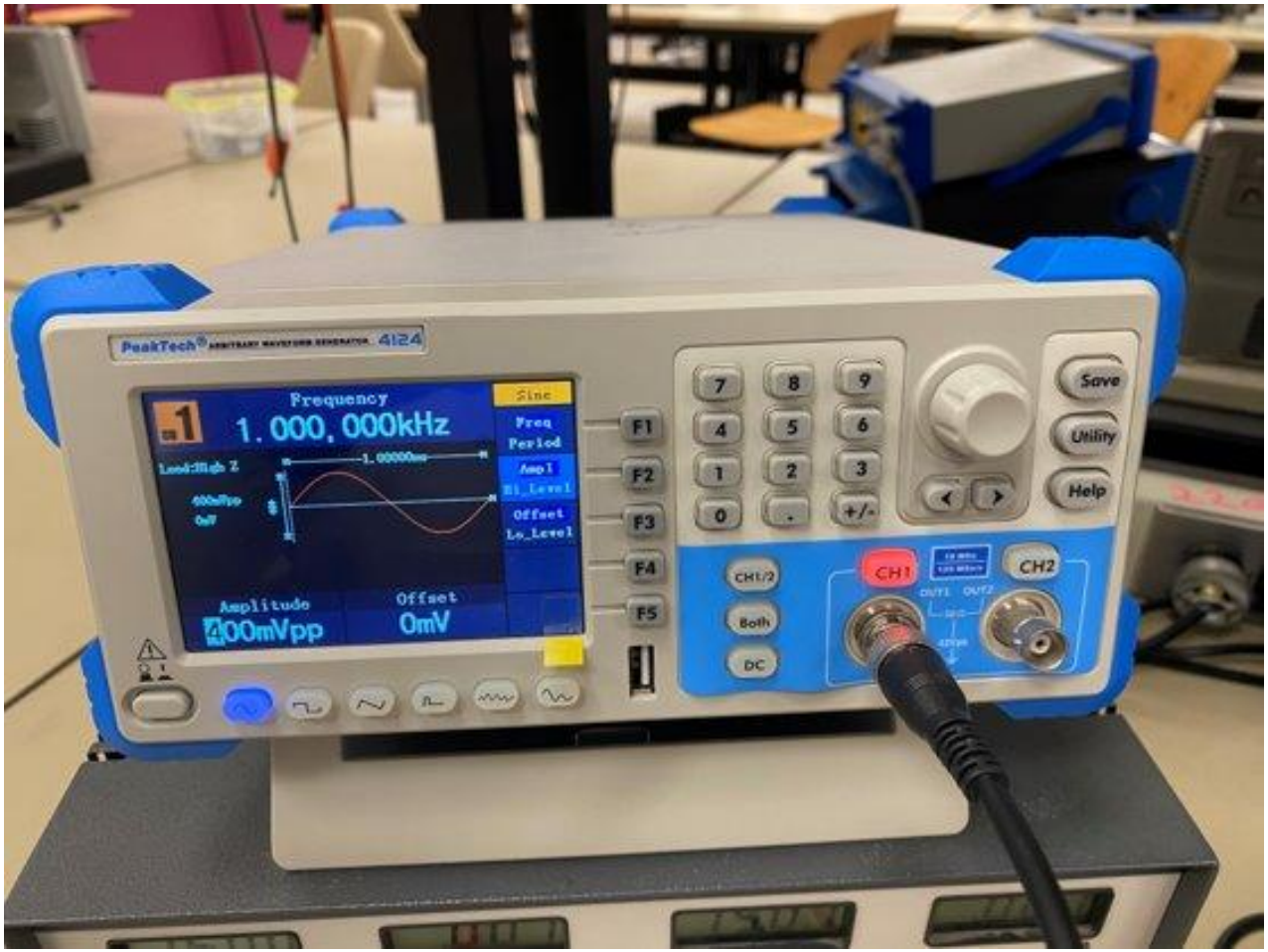


Noninverting Amplifier Circuit Connection (R_1 = 1 KΩ , R_2 = 10 KΩ)

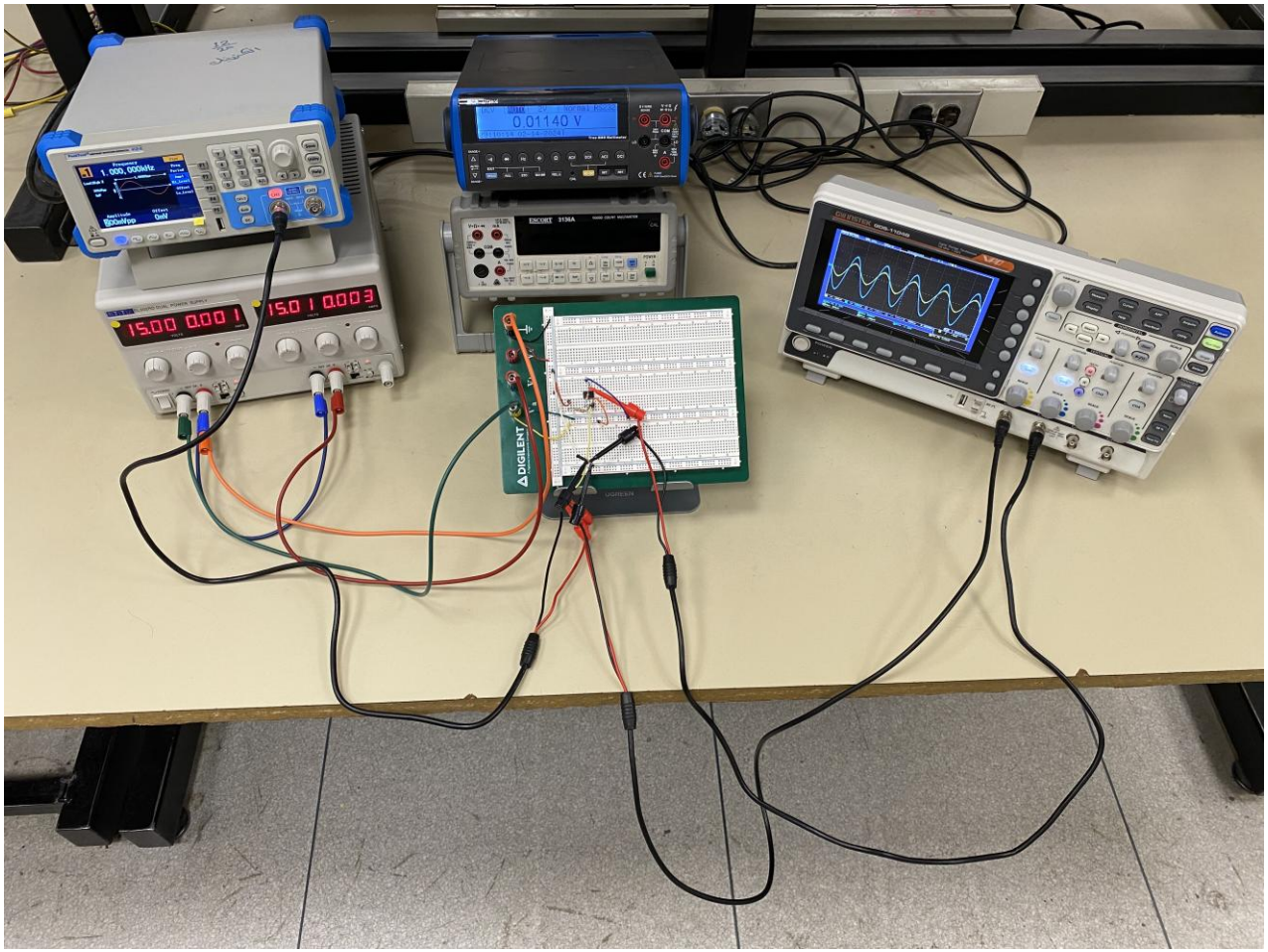




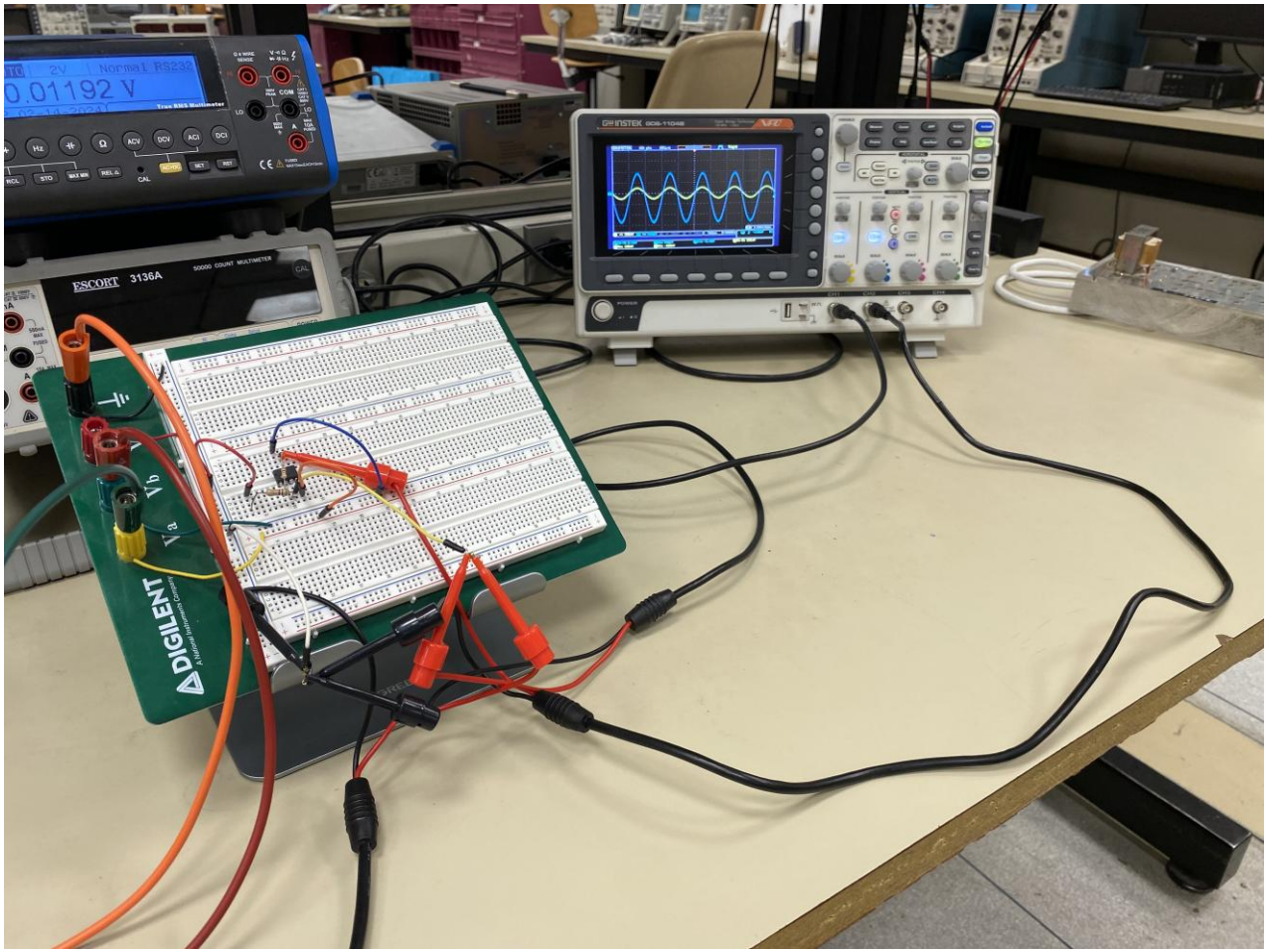
Waveform Generator Setting



Noninverting Amplifier Circuit Setup



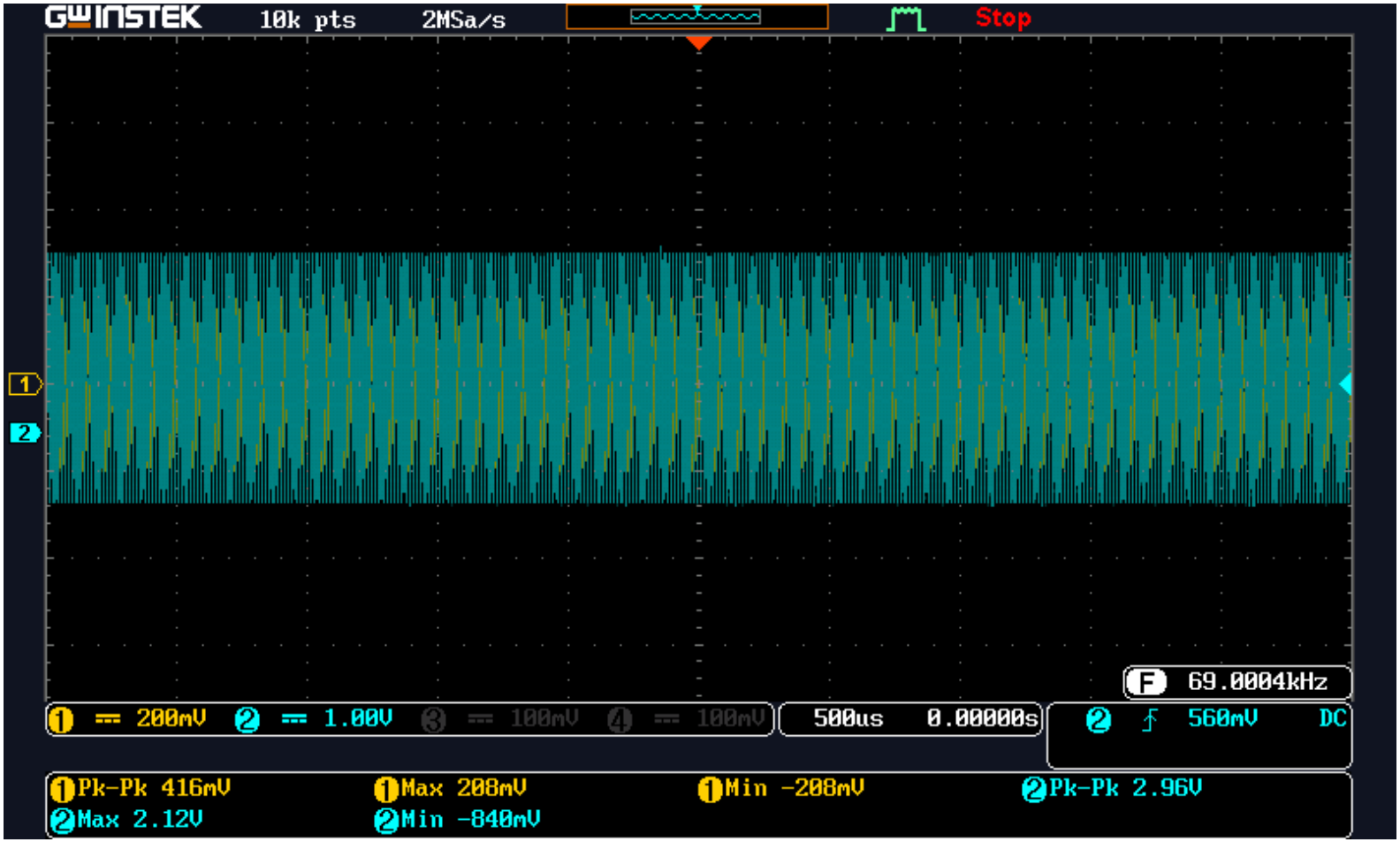
Circuit with External Connections



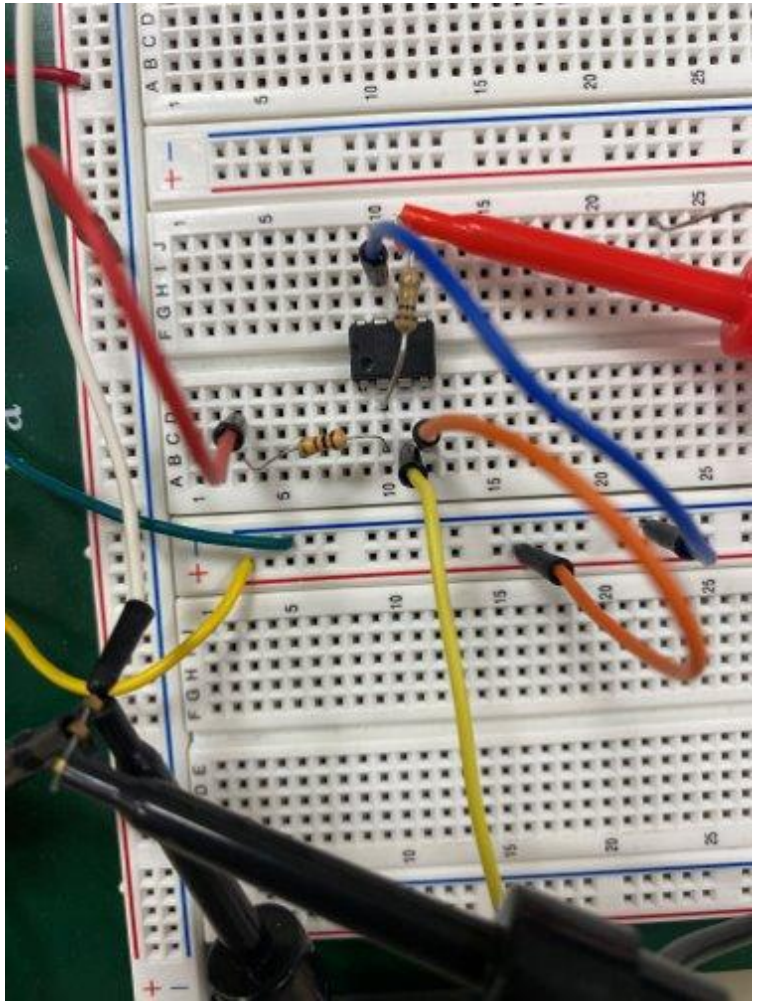
Noninverting Amplification (R_1 = 1 KΩ , R_2 = 10 KΩ)



Noninverting Amplifier Bandwidth



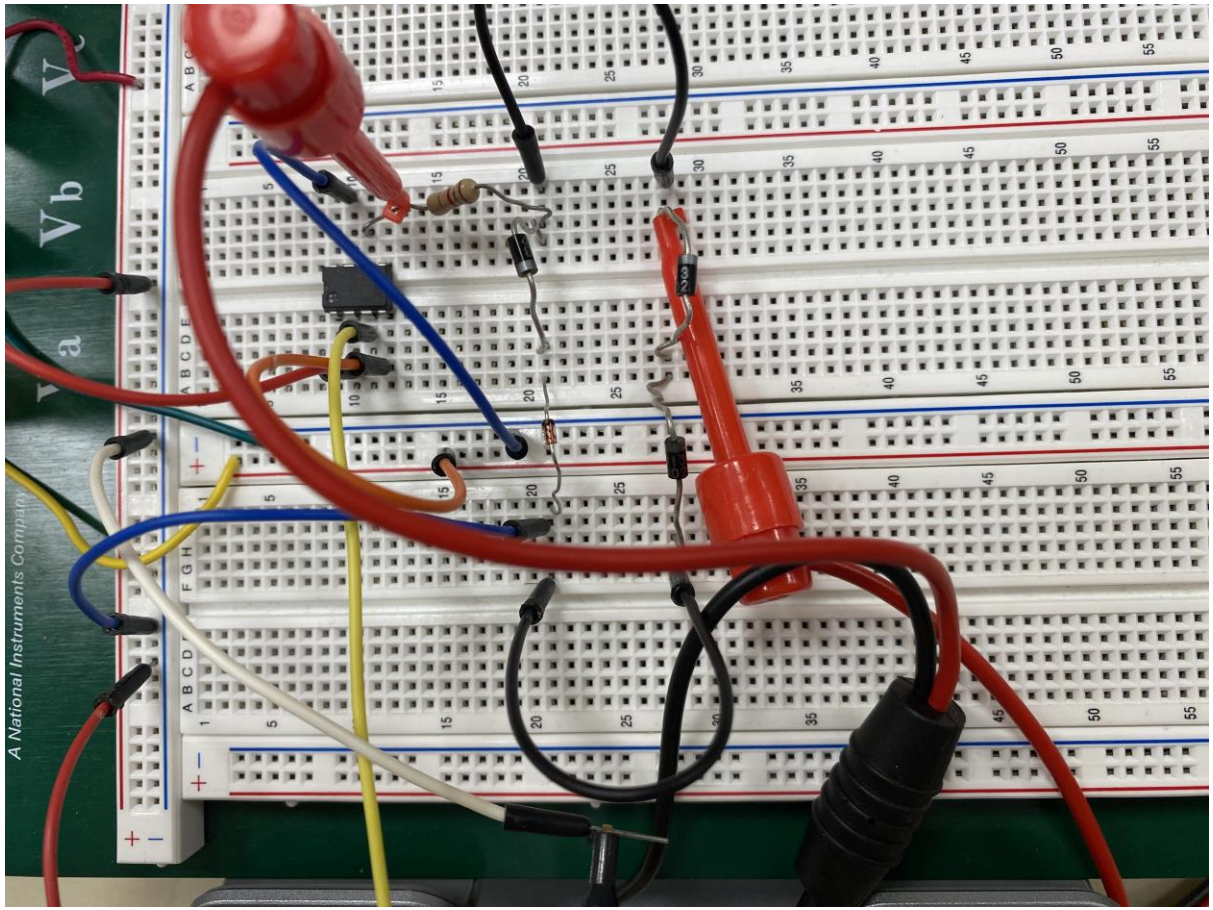
Noninverting Amplifier Circuit Connection (R_1 = 1 KΩ , R_2 = 100 KΩ)

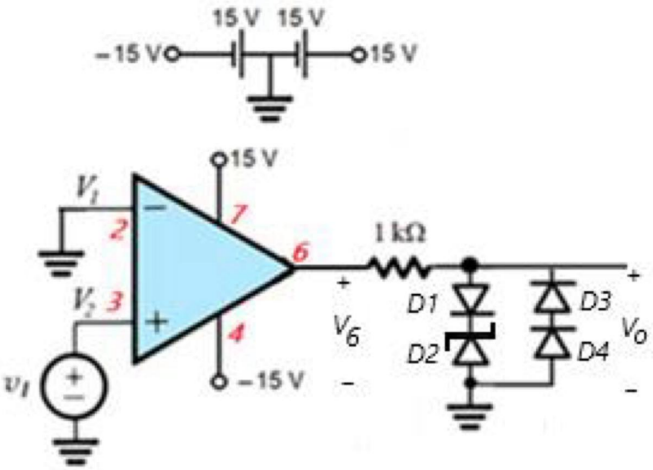


Noninverting Amplification (R_1 = 1 KΩ , R_2 = 100 KΩ)

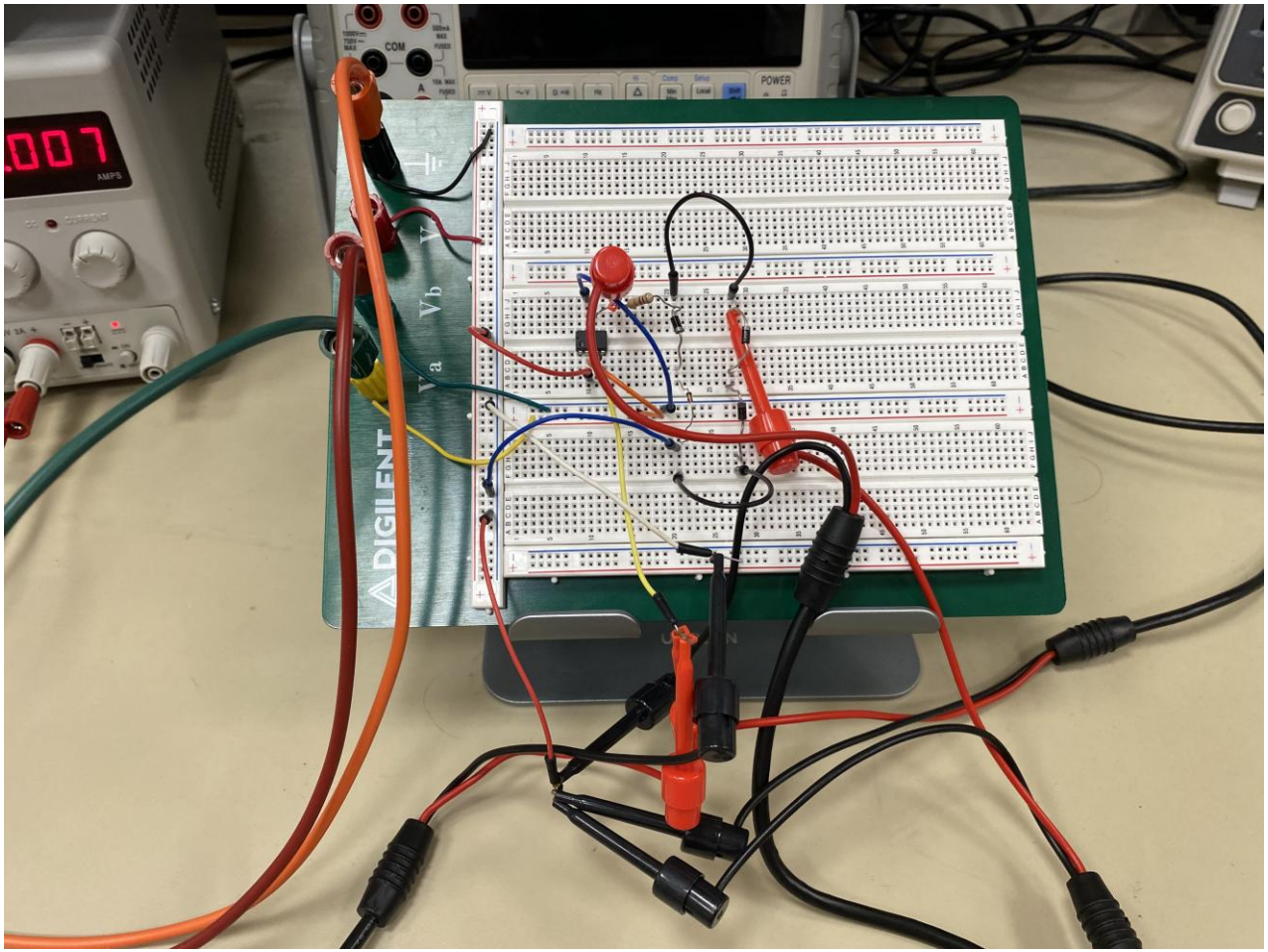


Comparator Circuit using Op Amp

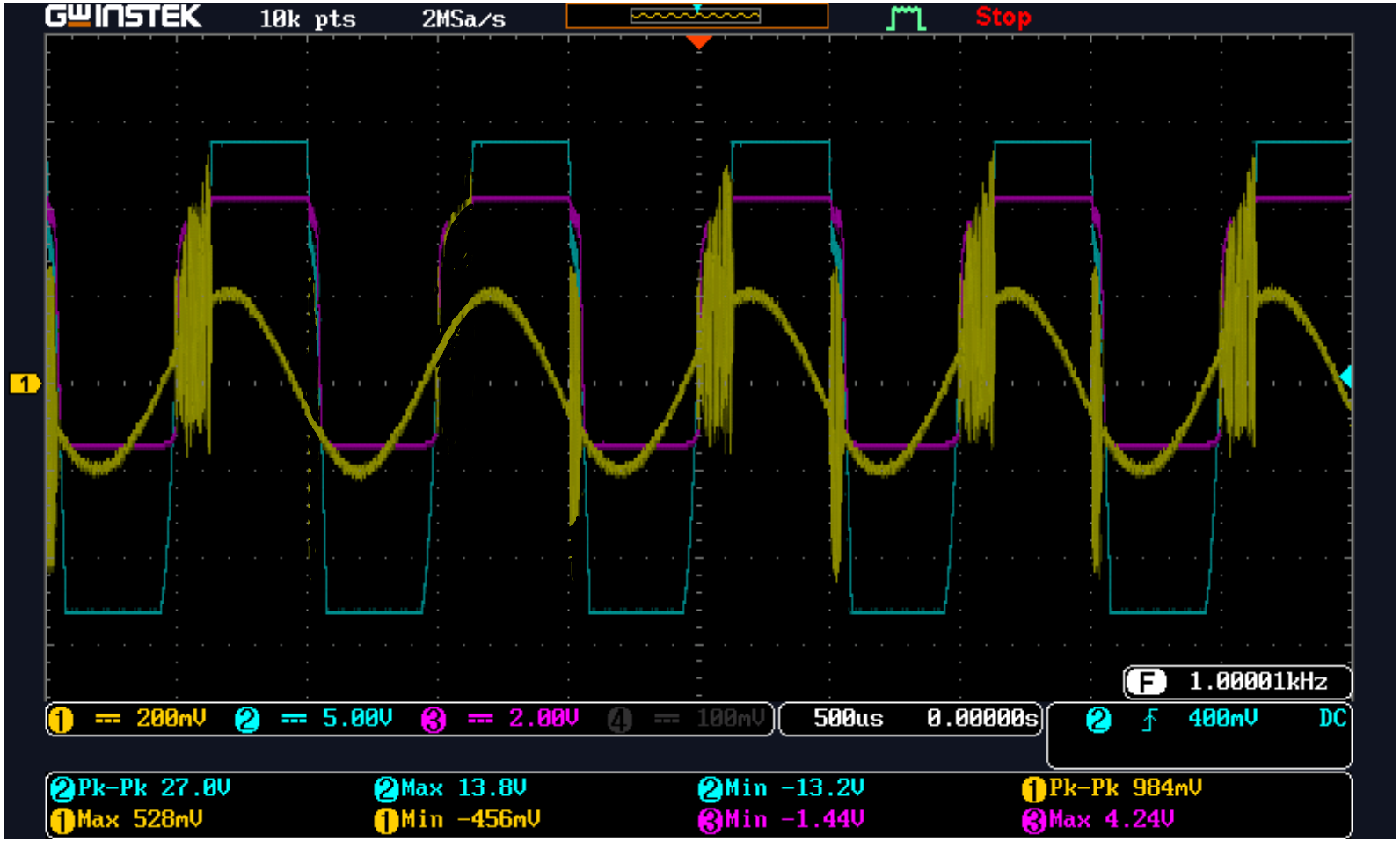




Comparator Circuit using Op Amp



Comparator Circuit using Op Amp



PROCEDURE A: Non-inverting Amplifier

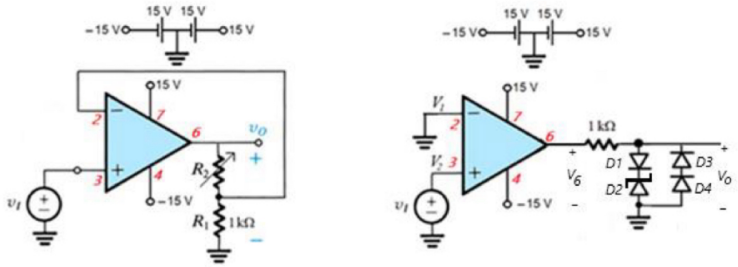


Figure 5.4 **(a)** The noninverting amplifier. **(b)** Comparator with limited output range.

1. Construct the circuit shown in Fig. 5.4a; start off with a value of 10 kΩ for R_2.
2. Using a waveform generator, apply a 1-kHz, ~~200~~ 400 $mV_{p\text{-}p}$ sinusoid to the noninverting terminal of your op amp.
3. Turn on your scope, wait for few seconds, press the **Default** key, and then the **Reset** key. Display $v_I(t)$ and $v_O(t)$ on **CH1** and **CH2** respectively.
4. Use the **SCALE** knob of **CH1** to set its vertical sensitivity at ~~100~~ 200 **mV/div**.
5. Use the **SCALE** knob of **CH2** to set its vertical sensitivity at ~~500 mV~~ 1 **V/div**.
6. Press the **Measure** key on your scope → **Add Measurement** → Activate the **Source 1** submenu → Use the **VARIABLE** knob to highlight **CH2** then press **Select** → Activate the **V/I** submenu → Highlight the appropriate item (**Pk-Pk, Max.** and **Min.**), then press the **Select** key to add anyone of them.
7. Switch the power supply, then waveform generator on.
8. Record $v_I(t)$ and $v_O(t)$ on Fig. 5.5.
9. Record the peak-to-peak amplitude for the input and output voltage waveforms, then calculate the voltage gain G: → max = 2.84 V

$$G = \frac{V_{o(\text{pk-pk})}}{V_{i(\text{pk-pk})}} \qquad (5.2)$$

$$= \frac{4.56}{424 \times 10^{-3}} = 10.75 \text{ V/V}$$

$V_{new} = 0.7 \times 2.84 \approx 2\text{ V}$, BW = 69 kHz

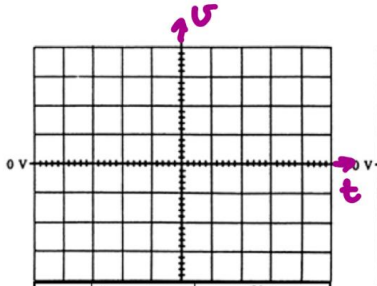
10. Increase the frequency of the input signal until the peak amplitude of the output falls to 0.7 of its value at 1 KHz. Take note of that frequency from the **waveform generator reading**. Record that frequency (the bandwidth BW) in Fig. 5.5.
11. Reduce the frequency of the input signal down to 1 kHz.
12. Increase the scale of **CH2** to **5 V/div**. Replace the 10-kΩ resistor with a 100-kΩ one, then sketch the resulting waveforms of $v_I(t)$ and $v_O(t)$ on Fig. 5.6.
13. Obtain the *upper limiting level* V_O^+ and the *lower limiting level* V_O^- for $v_O(t)$ from the scope Max. and Min. measurements. Record those values in Fig. 5.6.
14. Swich the waveform generator then the the power supply off.
    - How are these values related to the power rails ± 15 V?

PROCEDURE B: Implementing a Comparator Using an OP-AMP

1. Assemble the circuit shown in Fig. 5.4b.
2. Use the same input signal that you used for part A.
3. Connect the non-inverting input of the op amp to display to **CH1** of the scope, the op-amp output to **CH2**, and the overall circuit output to **CH3**.
4. Add a **Max.** and **Min.** measurement for **CH3** of the scope.
5. Set the the sacle of **CH3** at 2 V/Div.
6. Swich on the power supply t then the waveform generator on.
7. Record $v_I(t)$, $v_6(t)$, and $v_O(t)$ on Fig. 5.7.
8. Obtain the Max. and Min. values for the three waveforms from the oscilloscop and record them in lower part of Fig. 5.7.
9. Swich off the waveform generator then the the power supply off.

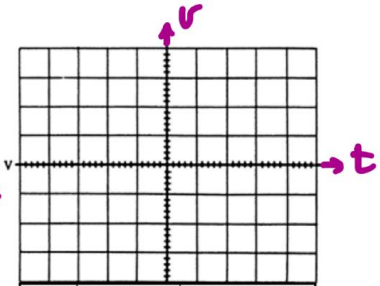
Time/div = 500μS/div

RESULTS



	v_I	v_O
Pk-Pk	424 mV	4.56 V
G =	10.75 V/V	
BW =	69 KHz	
G x BW =	741.75 KHz	

Figure 5.5

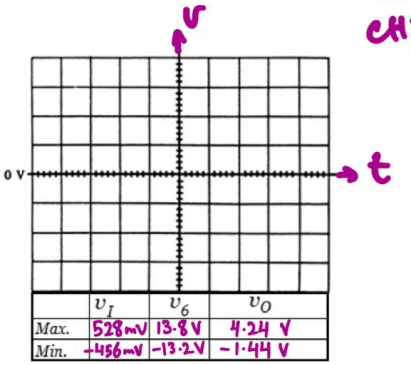


	v_I	v_O
Max.	280 mV	14.2 V
Min.	−216 mV	−14 V

Figure 5.6

CH1: 200mV/div
CH2: 1V/div

CH1: 200mV/div
CH2: 5V/div



	v_I	v_6	v_O
Max.	528mV	13.8V	4.24 V
Min.	−456mV	−13.2V	−1.44 V

Figure 5.7

CH1: 200 mV/div
CH2: 5 V/div
CH3: 2 V/div

UA741 Datasheet

Reference :
https://www.ti.com/lit/ds/symlink/ua741.pdf

µA741 General-Purpose Operational Amplifiers

1 Features

- Short-Circuit Protection
- Offset-Voltage Null Capability
- Large Common-Mode and Differential Voltage Ranges
- No Frequency Compensation Required
- No Latch-Up

2 Applications

- DVD Recorders and Players
- Pro Audio Mixers

3 Description

The µA741 device is a general-purpose operational amplifier featuring offset-voltage null capability.

The high common-mode input voltage range and the absence of latch-up make the amplifier ideal for voltage-follower applications. The device is short-circuit protected and the internal frequency compensation ensures stability without external components. A low-value potentiometer may be connected between the offset null inputs to null out the offset voltage as shown in Figure 12.

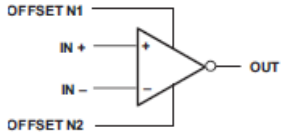
The µA741C device is characterized for operation from 0°C to 70°C.

Device Information(1)

PART NUMBER	PACKAGE	BODY SIZE (NOM)
µA741CD	SOIC (8)	4.90 mm × 3.91 mm
µA741CP	PDIP (8)	9.81 mm × 6.35 mm
µA741CPS	SO (8)	6.20 mm × 5.30 mm

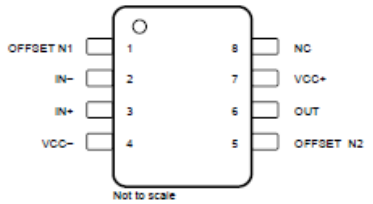
(1) For all available packages, see the orderable addendum at the end of the data sheet.

Simplified Schematic



5 Pin Configurations and Functions

uA741C D, P, or PS Package
8-Pin SOIC, PDIP, SO
Top View



Not to scale

NC- no internal connection

Pin Functions

PIN		I/O	DESCRIPTION
NAME	NO.		
IN+	3	I	Noninverting input
IN–	2	I	Inverting input
NC	8	—	No internal connection
OFFSET N1	1	I	External input offset voltage adjustment
OFFSET N2	5	I	External input offset voltage adjustment
OUT	6	O	Output
VCC+	7	—	Positive supply
VCC–	4	—	Negative supply

6 Specifications

6.1 Absolute Maximum Ratings

over virtual junction temperature range (unless otherwise noted)(1)

			MIN	MAX	UNIT
Supply voltage, V_{CC} (2)	μA741C		−18	18	V
Differential input voltage, V_{ID} (3)	μA741C		−15	15	V
Input voltage, V_I (any input) (2)(4)	μA741C		−15	15	V
Voltage between offset null (either OFFSET N1 or OFFSET N2) and V_{CC-}	μA741C		−15	15	V
Duration of output short circuit(5)			Unlimited		
Continuous total power dissipation			See *Thermal Information*		
Case temperature for 60 seconds		μA741C	N/A	N/A	°C
Lead temperature 1.6 mm (1/16 inch) from case for 60 seconds		μA741C	N/A	N/A	°C
Lead temperature 1.6 mm (1/16 inch) from case for 10 seconds	D, P, or PS package	μA741C		260	°C
Operating junction temperature, T_J				150	°C
Storage temperature range, T_{stg}		μA741C	−65	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) All voltage values, unless otherwise noted, are with respect to the midpoint between V_{CC+} and V_{CC-}.
(3) Differential voltages are at IN+ with respect to IN −.
(4) The magnitude of the input voltage must never exceed the magnitude of the supply voltage or 15 V, whichever is less.
(5) The output may be shorted to ground or either power supply.

6.2 Recommended Operating Conditions

			MIN	MAX	UNIT
V_{CC+}	Supply voltage		5	15	V
V_{CC-}	Supply voltage		−5	−15	V
T_A	Operating free-air temperature	μA741C	0	70	°C

6.3 Thermal Information

THERMAL METRIC(1)		μA741 D (SOIC) 8 PINS	μA741 P (PDIP) 8 PINS	μA741 PS (SO) 8 PINS	UNIT
$R_{\theta JA}$	Junction-to-ambient thermal resistance	129.2	87.4	119.7	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	73.6	89.3	66	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	72.4	64.4	70	°C/W
ψ_{JT}	Junction-to-top characterization parameter	25.9	49.8	27.2	°C/W
ψ_{JB}	Junction-to-board characterization parameter	71.7	64.1	69	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report.

6.4 Electrical Characteristics: μA741C

at specified virtual junction temperature, $V_{CC\pm}$ = ±15 V (unless otherwise noted)

PARAMETER		TEST CONDITIONS(1)		MIN	TYP	MAX	UNIT
V_{IO}	Input offset voltage	V_O = 0	25°C		1	6	mV
			Full range			7.5	
$\Delta V_{IO(adj)}$	Offset voltage adjust range	V_O = 0	25°C		±15		mV
I_{IO}	Input offset current	V_O = 0	25°C		20	200	nA
			Full range			300	
I_{IB}	Input bias current	V_O = 0	25°C		80	500	nA
			Full range			800	
V_{ICR}	Common-mode input voltage range	25°C		±12	±13		V
		Full range		±12			
V_{OM}	Maximum peak output voltage swing	R_L = 10 kΩ	25°C	±12	±14		V
		R_L ≥ 10 kΩ	Full range	±12			
		R_L = 2 kΩ	25°C	±10			
		R_L = 2 kΩ	Full range	±10			
A_{VD}	Large-signal differential voltage amplification	R_L ≥ 2 kΩ	25°C	20	200		V/mV
		V_O = ±10 V	Full range	15			
r_i	Input resistance	25°C		0.3	2		MΩ
r_o	Output resistance	V_O = 0; see(2)	25°C		75		Ω
C_i	Input capacitance	25°C			1.4		pF
CMRR	Common-mode rejection ratio	$V_{IC} = V_{ICRmin}$	25°C	70	90		dB
			Full range	70			
k_{SVS}	Supply voltage sensitivity ($\Delta V_{IO}/\Delta V_{CC}$)	V_{CC} = ±9 V to ±15 V	25°C		30	150	μV/V
			Full range			150	
I_{OS}	Short-circuit output current	25°C			±25	±40	mA
I_{CC}	Supply current	V_O = 0; no load	25°C		1.7	2.8	mA
			Full range			3.3	
P_D	Total power dissipation	V_O = 0; no load	25°C		50	85	mW
			Full range			100	

(1) All characteristics are measured under open-loop conditions with zero common-mode input voltage unless otherwise specified. Full range for the μA741C is 0°C to 70°C.
(2) This typical value applies only at frequencies above a few hundred hertz because of the effects of drift and thermal feedback.

Very high open-loop gain (A_{VD}=200000 V/V)
Very large input Impedance (r_i = 2 MΩ)
Very low output impedance (r_o = 75 Ω)
Very high common-mode rejection (CMRR = 90 dB)
Very high bandwidth (f_t = 1 MHz) see Fig.7

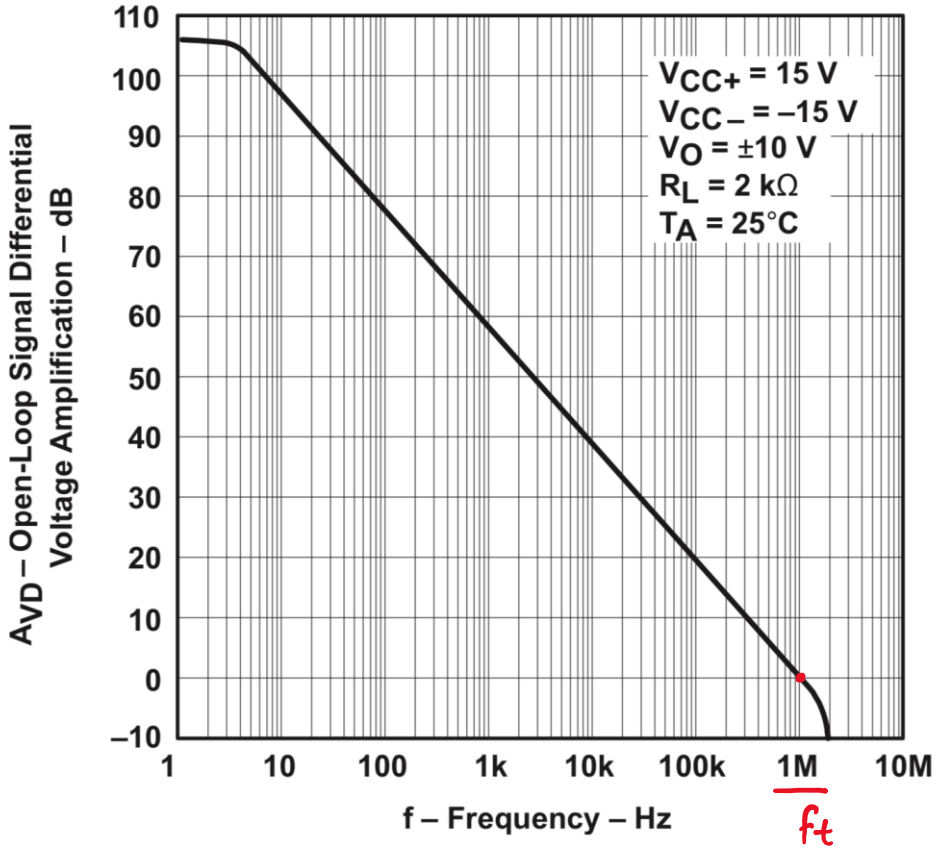


Figure 7. Open-Loop Large-Signal Differential Voltage Amplification vs Frequency